

## Adapting the Design

### 1. Frequent small rocks and debris: 10–15 mm, closely spaced

Here the main problem is that the foot can hit an obstacle during the swing phase. I would retune the **foot-side geometry, particularly the link that controls the vertical position/shape of the foot trajectory**, rather than simply increasing the whole mechanism size.

My first choice would be to slightly increase the effective length of the **foot-lifting link**  $h$ , while re-optimizing the neighboring foot triangle  $g - h - i$ .

The reason is that increasing this effective geometry can increase the clearance of the foot during the swing without requiring a completely different mechanism. The trade-off : increasing the swing height generally makes the foot travel through a larger vertical arc. That can increase:

- energy required to lift the leg,
- peak joint forces,
- swing time,
- and potentially reduce the useful horizontal stride during part of the swing.

It can also make the foot return to the ground at a slightly different angle.

So I would **not simply increase arbitrarily**. I would increase it slightly, regenerate the trajectory, and then adjust the neighboring foot-side pivot/link so that the stance portion remains approximately flat.

2. For smooth gravel, I would prioritize horizontal stride efficiency rather than additional clearance. I would retune the crank radius  $m$  and then re-optimize the stance-side linkage so that the foot spends a larger fraction of the cycle moving nearly horizontally. The trade-off is that increasing the crank radius can increase vertical excursion and actuator torque, so I would compensate by adjusting the adjacent linkage rather than simply maximizing  $m$ .
3. retune the **ground/stance geometry**, particularly the **fixed-pivot spacing and the crank-to-ground geometry**, so that the foot has a more favorable direction of motion during stance.

the fixed pivots determine the geometry of the four-bar loops. Changing their relative position changes the transmission angle and therefore the relationship between crank torque and the force transmitted to the foot.

- lower swing height
- reduced stride
- slower swing
- less symmetric trajectory